



Econometrics II

Fall 2025

Assignment 1: Forecasting consumer sentiment

Deadline: 5. of October, 23:59

1 Introduction

This paper utilises a univariate ARMA model to forecasts consumer sentiment from July 2022 to June 2025. The econometric ARMA theory is presented along with misspecification test statistics. Utilizing the General To Specific (GETS) approach - complementing with the Akaike Information Criterion (AIC) - we carry out model selection. The chosen model undergoes misspecification test, where there is indications of autoregressive conditional heteroskedasticity (ARCH) and non-normality of innovations errors. The model does not show surface level indications of autocorrelated error terms - which would be worrying for the consistency of the model. Comparing the forecast with realised observation we conclude that the model seems to have captured important structures in the data-process.

2 Presentation of data

The dataset consists of one series on US consumer sentiment published by the Federal reserve research database. The data ranges from January 1981 to July 2025. In figure 1 the log transformed series is presented. We use the log transformed variable because we are interested in autoregressive elasticities of consumer sentiment. Interestingly for our analysis we observe a sharp decline in 2022, which is where our sample used for fitting models will stop - and where we will perform a two year forecast of $\ln_csent = \log(UMCSENTx)$.

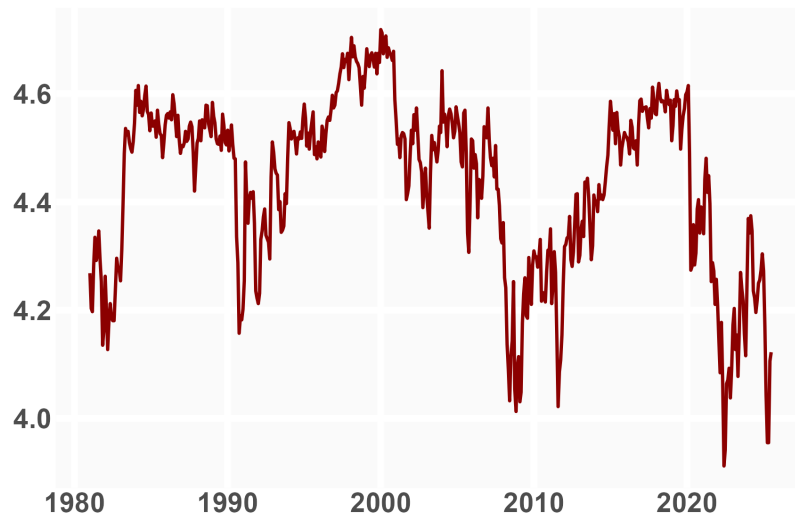


Figure 1. $\ln_csent$ over time

Note: The data used are from the assignment handout and is log transformed.

2.1 Graphical analysis of Autocorrelation functions

To get a sense of the datagenerating process' intertemporal correlation structure we estimate the Autocorrelationfunction (ACF) and the partial autocorrelation function (PACF). Figure 2 presents these estimations graphically.

Observing the left panel we conclude a relatively strong and long lasting time dependence. Although not exponentially declining, this is in large part in line with a theoretical ACF for a model with an autoregressive component. From the right panel we see that most of the time dependence seems to stem from the first lagged period. Although the estimation indicates that the third lag of `ln_csent` is statistically significant at the 5 pct. level. A starting point for our model selection will therefore be a model with around 3 autoregressive lags.

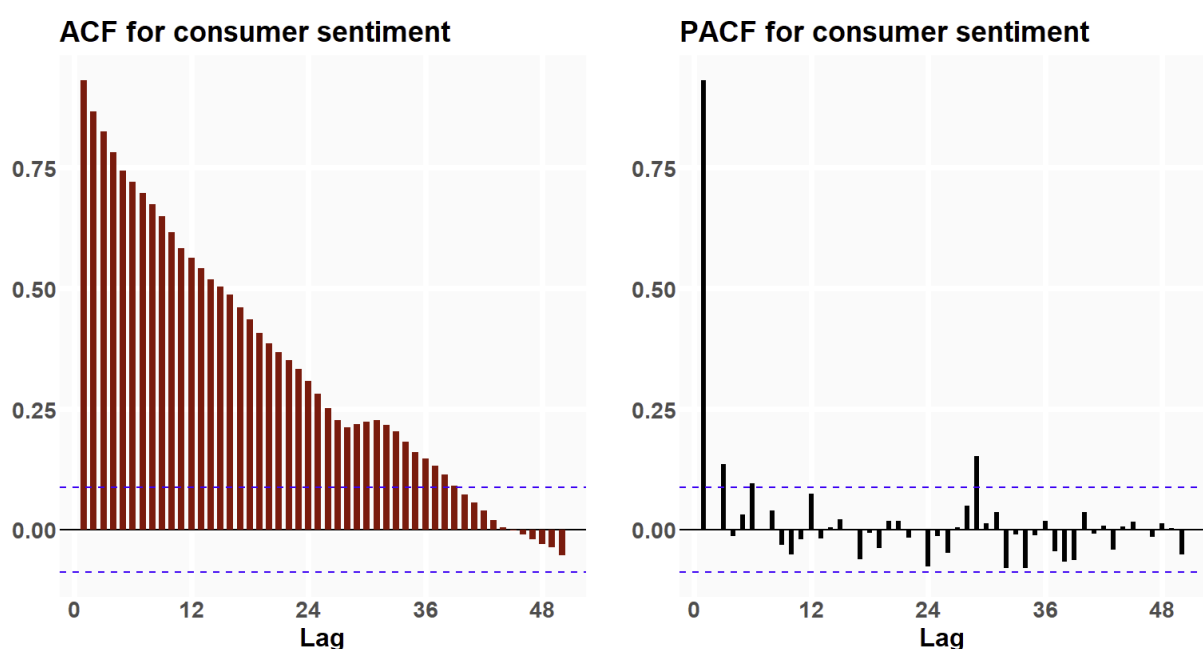


Figure 2: ACF and PACF for `ln_csent`

Note: The dashed lines show the 95pct. confidence bands for a standard normal distribution. These are useful because the estimated results will be asymptotically standard normally distributed (around the true value) if the timeseries `ln_csent` is stationary and weakly dependent. The estimation is carried out using the software R using the package `forecast` developed by Rob Hyndman.

3 Econometric theory

3.1 The ARMA model

This paper models the time series of consumer sentiment using an univariate autoregressive model. The general ARMA(p, q) model is given by

$$y_t = \delta + \epsilon_t + \sum_{i=1}^p \theta_i y_{t-i} + \sum_{i=1}^q \alpha_i \epsilon_{t-i}, \text{ where } \epsilon_t \text{ is i.i.d}(0, \sigma^2) \quad (3.1)$$

, where δ is a constant and $(0, \sigma^2)$ is an arbitrary i.i.d process $\{\epsilon_t\}$ characterised by a mean of 0 and a variance of σ^2 . This is often called a white noise process.¹ (3.1) is assumed to hold for observations $t = 1, 2, \dots, T$ with $y_0, y_{-1}, \dots, y_{-(p-1)}, \epsilon_0, \epsilon_1, \dots, \epsilon_{-(q-1)}$ given.

3.2 Maximum likelihood estimation

This paper estimates the parameters $\delta, \theta_1, \dots, \theta_p, \alpha_1, \dots, \alpha_q$ in (3.1) using the maximum likelihood estimator defined as

$$\hat{\theta}_{MLE} = \arg \max_{\theta \in \Theta} \log L(\theta | y_1, \dots, y_n) \quad (3.2)$$

where $\log L(\theta | y_1, \dots, y_n)$ is the log likelihood function. This estimation method required assuming an exact parametric datagenerating process and a specification of the parameter space.

Under four *high level assumptions*² it can be shown that the ML estimator is consistent

$$\hat{\theta}_{MLE} - \theta_0 \xrightarrow{p} 0 \quad (3.3)$$

and asymptotically normal distributed

$$\sqrt{n}(\hat{\theta} - \theta_0) \overset{a}{\sim} N(0, \Omega_\theta(\theta_0)) \quad (3.4)$$

, where $\Omega_\theta(\theta_0)$ is the inverse Fisher information: $\Omega_\theta(\theta_0) = \mathcal{I}^{-1}(\theta_0)$.

For all practical purposes we cannot hope to correctly assume the exact distribution characterising the actual data generating process. This, however, does not remedy the likelihood analysis useless. If we have that

$$E[s_t(\theta_0)] = 0 \quad (3.5)$$

where $s_t(\theta_0) = \nabla_\theta \ell(\theta) |_{\theta=\theta_0}$ is the score contribution vector of the stochastic variable y_t evaluated at the population parameter values θ_0 . Given (3.5) then $\mathcal{I}(\theta_0)^{-1}$ in (3.4) is replaced with the robust variance given by

$$\Omega_\theta^*(\theta_0) = \mathcal{I}(\theta_0)^{-1} \mathcal{J}(\theta_0) \mathcal{I}(\theta_0)^{-1} \quad (3.6)$$

In conclusion, if property in (3.5) holds but the distrubitive assumptions are not excacly as

¹Nielsen 2025, page 90

²Nielsen 2025, page 65

the true DGP then $\hat{\theta}_{QMLE}$ is the *quasi* maximum likelihood estimator, with the properties explained above.³

3.3 Stationarity and weak dependence

To estimate usefull parameters we need certain assumption on the underlying stochastic process to hold. Specifically weak stationarity and weak dependece will imply three of the four high level assumptions due to a LNN and a CLT.

A time series is weakly stationarity (or covariance stationarity) if for all t we have

$$E[y_t] = \mu, \quad V(y_t) = \gamma_0, \quad Cov(y_t, y_{t+h}) = \gamma_h, \text{ for } h \in \mathbb{N}$$

Weak dependence states that the stochastic vairables y_t and y_{t+h} become independent (sufficiently fast) as $h \rightarrow \infty$.

The time series in this empirical exercise represents a borderline case regarding stationarity. Following the assignment instructions, we assume that the series y_t is stationary in levels - and in logs.

3.4 Misspecification tests

When testing a given ARMA model we will focus on three test: Normality of error terms, ARCH and autocorrelation of error-terms.⁴ The tests are performed in R and we reject the null hypothesis if the p-value is less than 0.05.

The assumption of no-autocorrelation is important for the consistency of the estimator sinse we assume the DGP is an ARMA model and that

$$\mathbb{E}[\epsilon_t | \mathcal{I}_{t-1}] = 0 \implies 0 = \mathbb{E}[\epsilon_{t-h} \mathbb{E}[\epsilon_t | \mathcal{I}_{t-1}]] = \mathbb{E}[\mathbb{E}[\epsilon_{t-h} \epsilon_t | \mathcal{I}_{t-1}]] \stackrel{LIE}{=} \mathbb{E}[\epsilon_{t-h} \epsilon_t] \quad (3.7)$$

If this assumption is violated then our estimator will not generally be consistent - because we do not have conditional independence. This paper applies the Ljun-Box test statistic (or Partmantau), to test for no autocorrelation, given by

$$\xi_{LB} = T^2 \sum_{j=1}^h \frac{\hat{\rho}_j}{T-j} \xrightarrow{d} \chi^2(h-p) \text{ under } \mathcal{H}_0^5 \quad (3.8)$$

, where p is the number of ARMA parameters in the model being testet.

A Jaque-bera test is used for testing the assumption of nomally distributed errors. The null hypothesis $\mathcal{H}_0$ is that the residuals follow a normal distribution. Formally the

³Nielsen 2025, page 67

⁴Remember that we have assumed normality, no ARCH and uncorrelated error terms when constructing our likelihood function / joint probability density.

⁵Slideset on misspecification tests, slide 5

test tests the skew and kurtosis of the residuals which would be 0 and 3 respectively for a normal distribution and $\xi_{JB} \xrightarrow{d} \chi^2(2)$ under $\mathcal{H}_0$.⁶ If this is not fulfilled the ML-estimator is still consistent and asymptotically normally distributed if (3.5) holds.

To test for no-ARCH (autoregressive conditional heteroskedasticity) we make use of the auxiliary regression

$$\hat{\epsilon}_t^2 = \gamma_0 + \gamma_1 \hat{\epsilon}_{t-1}^2 + \dots + \gamma_h \hat{\epsilon}_{t-h}^2 + u_t \quad (3.9)$$

with the null hypothesis $\mathcal{H}_I : \gamma_1 = \gamma_2 = \dots = \gamma_h = 0$ and the alternative is volatility clustering, $\mathcal{H}_A : \exists \gamma_j \neq 0, i = 1, 2, \dots, h$. The ARCH test statistic has the asymptotic property $\xi_{ARCH} \xrightarrow{d} \chi^2(h)$ under $\mathcal{H}_0$.⁷

4 Empirical results

Our graphical analysis and discussion in section 2 points to a model with 3 autoregressive lags, and possibly some moving average terms. This paper considers the general to specific approach (GETS) as the main model selection tool. If a larger model nests the true model (of the data generating process) in its parameter space, then the MLE is still a consistent estimator (see section 3), and we can remove the statistically insignificant terms to simplify our model.

When modeling an ARMA process for empirical estimation, one should always be wary of canceling roots, which the estimated roots can be indicative of. Having this in mind we will include a relative modest number of autoregressive terms in our general model.

4.1 Model selection

Our approach when selecting the model will be to compare the ARMA(4,4) model with all nested models using an LR test. The chosen models will be the simplest AR(p) model and ARMA(p,q) model that we cannot reject being statistically equivalent at the 5 pct. level.

This results in two candidate models; AR(3) and ARMA(1,2). From tabel 1 we observe that the ARMA(1,2) model has the statistically insignificant estimate $\hat{\alpha}_1$, and therefore we prefer the AR(3) model. Note that the Aikon infomration criterion value is almost identical for the two - which much like the LR test is a measure of statistical similarity.

Tabel 1 also displays two other models. First we observe that we will reject the null hypothesis that AR(3) and AR(1) are statistically equivalent.⁸ In addition to this the ARMA(2,3) estimates are presented. This is the ARMA model that minimizes the AIC (and AICc) value - which is a different (and complementary) model selection approach

⁶Nielsen (2025), page 44

⁷Slideset on misspecification tests, slide 7

⁸With an Likelihood-ratio-test statistic of $-2(794.51 - 799.9) = 10.78 > F_{\chi^2(2)}(0.95) \approx 6$

to GETS. However the AR(2,3) model is not preferred to the AR(3) model, because of the insignificant estimates (and border line significant estimates) and due to a strong inclination to choose simpler models - as the assignment instructions also reminds us.

	AR(1)	AR(3)	ARMA(1,2)	ARMA(2,3)
AR-1	0.96*** (0.01)	1.00*** (0.04)	0.97*** (0.01)	0.11 (0.11)
AR-2		-0.18** (0.06)		0.84*** (0.10)
AR-3		0.14** (0.05)		
MA-1			0.02 (0.05)	0.90*** (0.11)
MA-2			-0.14** (0.05)	-0.14* (0.07)
MA-3				-0.20*** (0.05)
Intercept	4.42*** (0.05)	4.42*** (0.06)	4.42*** (0.06)	4.42*** (0.06)
Jaque-Bera	0.0	0.0	0.0	0.0
Ljung-Box	0.0867	0.2574	0.2570	0.5838
ARCH	0.0021	0.0029	0.0023	0.0022
AICc	-1582.97	-1589.69	-1589.25	-1592.02
Log-Lik	794.51	799.90	799.69	803.12

Tabel 1: ARMA Model Estimation Results

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Estimated standard errors are in paranthesis. These are calculated as the inverse information, and is therefore not robust to ARCH or non-normal distributed innovations

The model of choice therefore is the AR(3)-model, where the Ljung-Box test indicates no remaining autocorrelation, satisfying the white-noise assumption for ε_t . However, the ARCH test reject the Null-Hyp. of zero homoskedasticity and Jarque-Bera rejects the Null-Hyp. of normality. Hence, the MLE estimates are interpreted as QMLE, which remain consistent and asymptotically normal under correct mean specification, allowing for valid inference with robust standard errors.

Parameter	Robust std. err.	P-value
AR-1	0.056	0.000
AR-2	0.073	0.025
AR-3	0.048	0.009
Intercept	0.063	0.000

Tabel 2: AR(3) with Quasi-maximum likelihood covariance matrix used for robustness

Table 2 reports the robust (QMLE) standard errors for the AR(3) model. All coefficients remain statistically significant at a 5% significance level. Confirming the persistence and cyclical dynamics found previously. The use of robust standard errors corrects for potential heteroskedasticity and non-normality in the residuals, ensuring valid inference even when the classical MLE assumptions are violated.

4.2 Forecasting with ARMA

A forecast is the best prediction of future observations given our model and the information set $\mathcal{I}_t$ containing all realised values of the stochastic process $\{y_t\}$. We calculate the equations for the first three forecasted periods.

$$\begin{aligned}\mathbb{E}[y_{t+1}|\mathcal{I}_t] &= \delta + \sum_{i=1}^p \theta_i y_{(t+1)-i} + \sum_{i=1}^q \alpha_i \epsilon_{(t+1)-i} \\ \mathbb{E}[y_{t+2}|\mathcal{I}_t] &= \delta + \theta_1 \mathbb{E}[y_{t+1}|\mathcal{I}_t] + \sum_{i=2}^p \theta_i y_{(t+2)-i} + \sum_{i=2}^q \alpha_i \epsilon_{(t+2)-i} \\ \mathbb{E}[y_{t+3}|\mathcal{I}_t] &= \delta + \sum_{i=1}^2 \theta_i \mathbb{E}[y_{(t+3)-i}|\mathcal{I}_t] + \sum_{i=3}^p \theta_i y_{(t+3)-i} + \sum_{i=3}^q \alpha_i \epsilon_{(t+3)-i} \\ &\vdots\end{aligned}$$

In practice, the forecast is obtained by inserting the estimated parameters $\hat{\theta}_1, \dots, \hat{\theta}_p$ and the most recent observed values of y_t into the AR(3) model. The past residuals $\hat{\epsilon}_t$ are used to update the moving average terms when applicable. Based on these estimates, the expected squared forecast error, also known as the Forecast Error Variance (FEV), is computed as

$$\text{FEV}(h) = \mathbb{E}\left[\left(y_{t+h} - \mathbb{E}[y_{t+h}|\mathcal{I}_t]\right)^2 \middle| \mathcal{I}_t\right]$$

which measures the uncertainty of the forecast h periods ahead. In the empirical application, the FEV is approximated by the sample variance of the forecast residuals over the evaluation period (2022(7)–2025(7)).

Figure 3 presents the dynamic forecast from the AR(3) model for the period 2022(7) – 2025(7), together with the actual log-consumer-sentiment index. The model captures the strong decline in sentiment during 2022 and predicts a gradual recovery towards its long-run mean. The widening confidence interval reflects the increase in forecast uncertainty, consistent with the rise in the forecast error variance (FEV) as the forecast horizon extends.

To assess forecast accuracy, standard forecast-error metrics are reported. The root mean squared error (RMSE = 0.081) and mean absolute error (MAE = 0.070). These indicate that the AR(3) model's predictions deviate only modestly from the observed values. Overall, the model provides a reliable short- to medium-term forecast of consu-

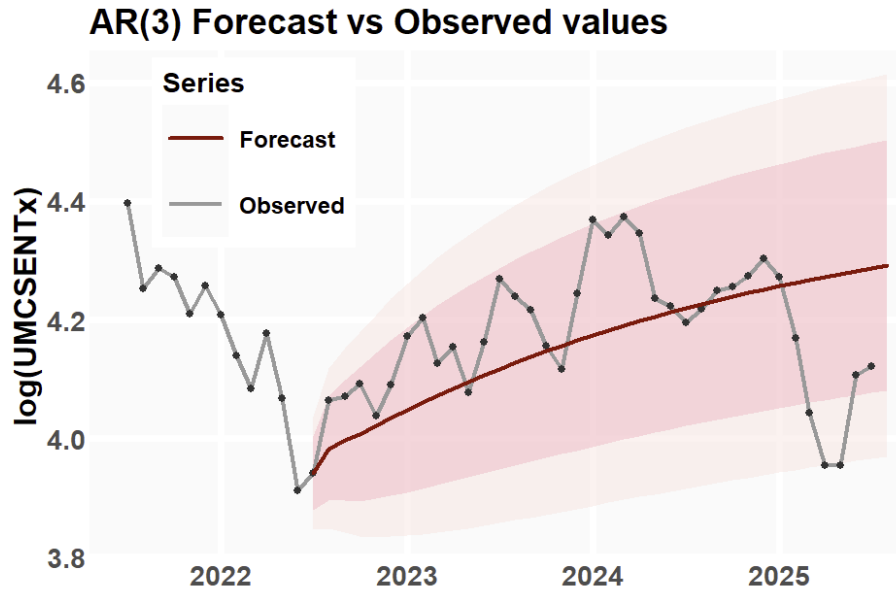


Figure 3. forecast AR(3)

mer sentiment. The predicted slow recovery after mid-2022 is broadly consistent with the observed data, although short-term volatility is slightly underestimated. The results therefore suggest that consumer sentiment adjusts gradually to shocks, reflecting the persistent dynamics captured by the autoregressive terms.

5 Discussion and conclusion

This paper examined the dynamics of U.S consumer sentiment using an ARMA framework. Utilizing the GETS approach, and assuming stationarity of the data, we found the AR(3) (ARMA(3,0)) model to be the best consistent estimator. Diagnostic testing showed no remaining autocorrelation, but heteroskedasticity and non-normality remained as issues for valid inference. These were addressed by using robust std. err. which ensured consistent inference under QMLE. All autoregressive coefficients remained statistically significant.

The estimated autoregressive coefficients from the AR(3)-model indicate a high degree of persistence in consumer sentiment. Specifically, the large and statistically significant AR(1)-term close to unity implies that shocks to confidence tend to have long-lasting effects, with sentiment adjusting only slowly over time. The smaller negative AR(2) and positive AR(3) coefficients suggest mild cyclical dynamics, meaning that consumer confidence exhibits small oscillations before gradually returning to its steady state level.

The model parameters are later examined to test the predictive performance of them. The forecasting exercise demonstrated that the AR(3) model provides a relatively accurate short to medium term prediction, when predicting a steady recovery towards the long run mean after the shock in 2021-2022.